

Ultrafast Noise-Immune Short-Circuit Detection in Gallium Nitride Power Semiconductors via Drain-Source Voltage Pattern Recognition

Tian Qiu

Electrical, Computer, and System
Engineering
Rensselaer Polytechnic Institute
Troy, USA
qiu2@rpi.edu

Zheyu Zhang

Electrical, Computer, and System
Engineering
Rensselaer Polytechnic Institute
Troy, USA
zhangz49@rpi.edu

Annoy Kumar Das

Electrical, Computer, and System
Engineering
Rensselaer Polytechnic Institute
Troy, USA
dasa7@rpi.edu

Israel Zikpi

Electrical, Computer, and System
Engineering
Rensselaer Polytechnic Institute
Troy, USA
zikpii@rpi.edu

Abstract—As gallium nitride (GaN) devices prevail in power electronics, the operational safety and reliability attract more attention. Due to the small thermal mass and limited short-circuit (SC) withstand capability, the SC protection for GaN devices is crucial. One widely accepted scheme is desaturation (Desat) protection based on ON-state drain-source voltage (v_{ds}) information, which, however, faces a design trade-off between response time and noise immunity, leaving its feasibility questionable for GaN applications. This paper proposes a novel SC detection scheme that leverages the pattern difference of the entire v_{ds} between normal operation and fault scenarios, in terms of switching, ON- and OFF-states, breaking the compromise and achieving the targets of ultrafast detection with noise immunity. First, the characteristics of fault patterns are investigated, and detection strategy is accordingly discussed to distinguish the fault from normal operation. Then, the capacitor divider design is proposed for capturing the critical v_{ds} patterns with noise immunity. Based on the detection strategy, the logic synthesis is formulated to generate the fault flag via digital operation on v_{ds} patterns. Finally, circuit simulation and hardware tests are performed for design validation. The ultrafast SC detection time are observed of around 30 – 40 ns, which is actually even smaller excluding the propagation delay of isolators in the measurement.

Keywords—Short circuit protection, GaN HEMT, short-circuit withstand time, ultrafast detection, voltage pattern

I. INTRODUCTION

Gallium nitride (GaN) power semiconductors have heightened the need for fast and robust overcurrent and short-circuit protection schemes to ensure converter reliability. On the one hand, GaN exhibits exceptional material properties, such as higher breakdown electric fields and saturation velocities, allowing a smaller die size and thereby reducing the junction parasitics to improve the switching speed [1]. On the other hand, however, the small die/package dimension makes it more challenging for heat dissipation, particularly during the short-circuit (SC) event. Compared to Si and SiC, GaN has smaller short-circuit withstand time (SCWT). As reported in [2], applying a voltage over 400 V reduces the SCWT of a 650-V GaN HEMT within 600 ns, which is way more critical than Si devices ($\sim 10 \mu\text{s}$). In [3], the simulation indicates much shorter SCWT of GaN at higher ambient temperature, as shown in Fig. 1, requiring the SC protection response time within hundreds of nanoseconds or even faster.

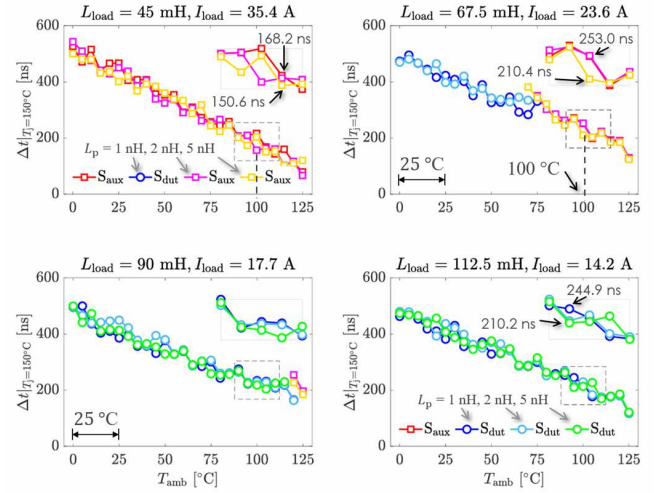


Fig. 1. SCWT of GaN evaluated under the conjoint effects of ambient temperature, loop parasitics and load current. SCWT is defined as the time taken for junction temperature to reach 150 °C under sustained SC fault. DUT is GS66516T [4] by GaN Systems/Infineon; details of the parametric sweep analysis can be found in [3].

One well-accepted SC protection for wide bandgap (WBG) power semiconductor devices is the desaturation (Desat) protection, which can proceed with small response time down to 100 ns [5]. The inherent advantage of this method is the direct availability of the necessary signal, i.e., the drain-source voltage v_{ds} being an easy to process signal. However, due to high dv/dt associated with GaN, oscillation/ringing and percentage overshoot in the drain-source voltage due to loop parasitic become more severe, leading to a compromise in determining the necessary blanking time t_{blk} and detection threshold voltage $v_{SC,th}$ for implementing the Desat protection reliably. For example, a simple desaturation-based protection circuit for GaN HEMT with ultrafast response is demonstrated in [5] based on a selection of $v_{SC,th} = 3 \text{ V}$. However, on account of the presence of loop inductance, sustained oscillation is present in the drain-source voltage even during normal turn ON condition, thus demanding higher margin for t_{blk} in order to avoid false triggering until the v_{ds} amplitude becomes lower than $v_{SC,th}$. For instance, it is reported in [3] that introduction of only 1 nH of additional loop inductance lead to 227% increase in t_{blk} to

avoid false triggering during normal turn-on. As t_{blk} is increased to counter the effect of higher oscillation due to loop parasitic, this inhibits fast/immediate detection of SC fault using Desat protection and allows the fault current to increase until t_{blk} is expired, leading to higher current stress, loss and temperature rise in the GaN device. In addition, GaN has positive temperature-dependent ON-state resistance $R_{\text{ds(on)}}$, which may elevate the desired protection threshold in configuration to avoid mis-triggering at high operating temperature. As we take into account the cumulative effects of loop parasitic, junction temperature and load current, it becomes apparent that the SCWT of GaN decreases monotonically as illustrated in Fig. 1. This evidences that SCWT becoming smaller at combinations of elevated junction/ambient temperature, higher load and larger loop parasitic will require a faster protection, conflicting with the elevated threshold in the case of conventional Desat protection. On top of that, the unique behavior of dynamic $R_{\text{ds(on)}}$ makes the threshold design coupled with other factors such as bus voltage, switching mode, etc. [3]. Overall, there is a trade-off between response time and noise immunity, which challenges the feasibility of Desat protection for GaN.

In addition to Desat protection, several detection methods have been proposed in literature, such as gate charge-based and i_d -based schemes [6]. Generally, the gate-based scheme obviates the blanking time, but it's not applicable for fault-under-load (FUL) type SC. Besides the demanding calculation of the gate charge (considering runtime and drift of amplifiers, especially during FUL), using gate charge-based method with SiC MOSFETs is problematic, since the miller capacitance C_{gd} is very small, making it extremely challenging to distinguish between a hard-switching fault (HSF) and a normal turn-on operation [7], [8]. The i_d -based direct detection schemes intuitively detect the overcurrent/short-circuit issue but have drawbacks regarding the current sensors. Using a current shunt to directly measure i_d is the most straightforward method [9]. However, besides not being galvanically isolated, the resistance of the shunt, being in the same range as that of the drain-source resistance of device, leads to additional conduction losses, which cannot be neglected. Furthermore, the additional stray inductance introduced in the commutation loop due to the shunt, can significantly deteriorate the switching performance of WBG devices. The current transformer is bulky, posing challenges for integration into the driver or power loop [10]. The on-chip current sensor can be integrated on the chip thus detecting the short-circuit current accurately, but it reduces the effective area of chip active area [11]. The Rogowski coil can be integrated in the power circuit board, and has high noise immunity [12], [13], [14]. Rogowski coil-based i_d detection method can respond to SC event very quickly under any operating condition. However, the use of Rogowski coil brings complexity of circuit design and the implementation cost is relatively high. Moreover, the fast switching and small SCWT

of GaN challenge the compatibility of these i_d - or gate charge-based approaches reported in the prior studies.

In this paper, we propose a new SC detection scheme. Unlike relying on merely the magnitude of v_{ds} in Desat, it leverages pattern difference in the entire v_{ds} between normal operation and SC faults, accounting for fault occurrence at switching, ON- and OFF-states, to achieve ultrafast detection and noise-immunity simultaneously. In Section II, the fault pattern characteristics and detection strategy are discussed to establish the conception of v_{ds} -pattern-based fault recognition. In Section III, the design consideration of the detection circuit and operational logic synthesis are proposed based on the analysis in Section II. In Section IV, the SC test in phase-leg configuration is performed with LTspice simulation and hardware prototype to validate the design function and detection response speed.

II. BASIC CONCEPT OF FAULT PATTERN RECOGNITION FOR SHORT-CIRCUIT DETECTION

A. Fault Pattern Characteristics

In conventional Desat protection, the magnitude of v_{ds} in the saturation region is utilized as the key input to generate the fault flag, the detection speed of which largely relies on the configuration of blanking time and detection threshold. The concept of fault-pattern detection presented in this paper utilizes not only the magnitude, but the entire information of v_{ds} including the slew-rate, and the sign of slew-rate throughout every moment (switching, ON- and OFF-states) to recognize the SC fault while bypassing the normal operation.

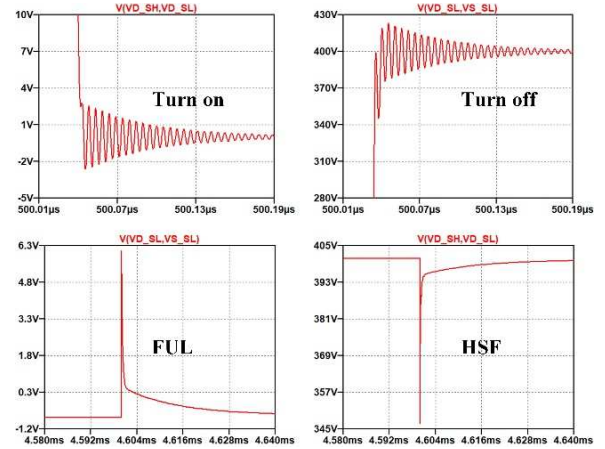


Fig. 2. Distinct patterns in v_{ds} among different scenarios.

Fig. 2 illustrates the simulated waveforms regarding a set of typical drain-source voltage v_{ds} at normal turn-on, -off, HSF and FUL. Apparently, amplitude, slew-rate, and the sign of slew-rate of v_{ds} are quite different during switching, ON- and OFF-states between the normal operation and SC fault. Based

TABLE I CHARACTERISTICS IN v_{ds} PATTERNS AT NORMAL OPERATION AND FAULT (650-V GaN WITH 400-V DC BUS AS AN EXAMPLE).

	Turn-on transition	Turn-off transition	ON state		OFF state	
			Normal	Fault	Normal	Fault
Δv_{ds} [V]	~ 400	~ 400	1 – 10	1 – 10	30 – 40	40 – 50
dv/dt [V/ns]	> 40	> 40	< 2	2 – 4	< 10	20 – 40
Sign	-	+	Alternating + & -	+ / -	Alternating + & -	- / +

on the presented case study in Fig. 2, the characteristics of v_{ds} patterns are summarized in TABLE 1. The key difference between normal operation and SC scenarios attributes to:

- 1) Δv_{ds} at normal ON/OFF-states is basically induced by i) switching noise and ii) fluctuation in DC bus voltage. At switching transition, Δv_{ds} is equal to the DC bus voltage and is much larger than that at ON/OFF-states. However, according to TABLE 1, the magnitude of Δv_{ds} at either ON/OFF-states or SC scenarios is pretty similar, which leaves potential risk of mis-triggering the detection.
- 2) dv/dt at either switching transition or SC scenarios has similar magnitude and is much larger than that at ON/OFF-states.
- 3) The sign of dv/dt is negative at i) turn-on transition and ii) HSF scenarios, while it becomes positive at i) turn-off transition and ii) FUL scenarios. At ON/OFF-states, the sign of dv/dt is alternately positive and negative due to the periodic Δv_{ds} .

Overall, at ON/OFF-states and SC scenarios, v_{ds} patterns are coupled in terms of Δv_{ds} magnitude, while at switching transition and SC scenarios, they are coupled in terms of dv/dt . To detect the SC faults with robustness against misdiagnosis, the key is accurately extracting the characteristics of v_{ds} and distinguish the fault and normal operation in a strategic way.

B. Fault Detection Strategy

In order to precisely extract and distinguish the fault pattern, a sensing circuit with a high bandwidth is required. In this paper, we propose a RC network, which is referred to as a “capacitor divider” in Section III, to swiftly scale down v_{ds} , thus making it compatible for fast signal processing. The advantage of the RC network is that instead of v_{ds} but Δv_{ds} will be captured regardless of its initial state. The reason, as indicated in TABLE 1, is that only the variation of v_{ds} matters in fault detection. With multiple RC networks of different sensitivity to dv/dt , the difference in v_{ds} between fault and normal operation could be captured and then identified via signal processing approach.

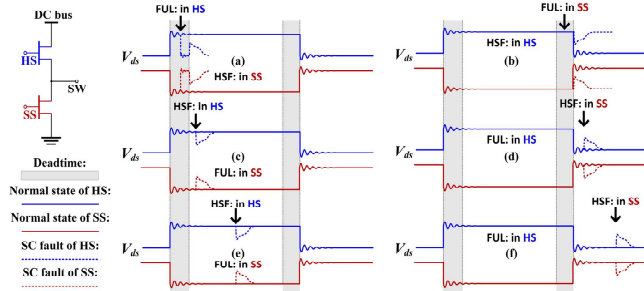


Fig. 3. Fault occurrence in v_{ds} throughout a period in a phase leg.

The fundamental idea of the signal processing for SC detection is: Once dv/dt is captured, whether it is “normal” or “fault” is judged based on its moment of occurrence. In a phase leg with PWM operation, the occurrence of SC fault basically includes six patterns, as shown in Fig. 3. The blue color represents hard-switching operation and the red color represents soft-switching operation. The black arrow indicates the moment of SC occurrence. Fault patterns (a) and (b) illustrate the SC fault

of respectively the hard- and soft-switching device occurring in the deadtime when both device are OFF, thus preventing the shoot-through happening until the deadtime ends. The SC fault, right after the deadtime, thereby becomes FUL for the hard-switching device and HSF for the soft-switching device in pattern (a), and HSF for the hard-switching device and FUL for the soft-switching device in pattern (b). In patterns (c) and (e), SC occurs in the hard-switching device at its OFF state, but in (c) the occurrence moment is next to the switching transition while in (e) the occurrence is located at stable OFF state. Similar situation for (d) and (f) where the fault happens in the soft-switching device. It is noted that a major difference between (c) and (d) is that the fault is not coupled with the switching noise in (c) because of the deadtime that decays the oscillation. However, it is coupled with the switching noise in (d), where the switching transition happens right after the deadtime due to the nature of synchronous switching.

Accordingly, a logic synthesis is proposed and achieved by a digital circuit developed in this paper, which will be delivered in detail in Section III. With the designed logical operation, the aforementioned six SC fault patterns could be recognized and distinguished from the normal operation. On top of that, one advantage of the digital circuit approach is the ultrafast speed and small propagation delay (e.g., several ns) with low cost and strong reliability.

III. DESIGN CONSIDERATIONS AND CIRCUIT IMPLEMENTATION

A. Sensing and Digitalization of Fault Patterns

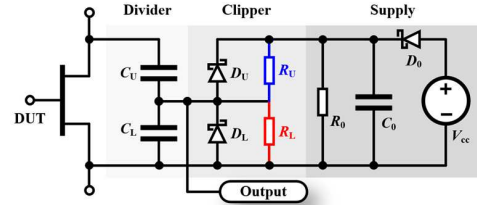


Fig. 4. Capacitor divider unit diagram.

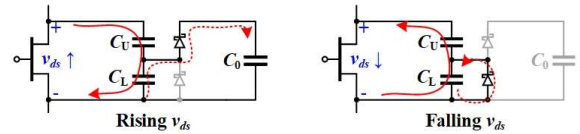


Fig. 5. Operation principle of the CDU.

The capacitor divider unit (CDU) is proposed in Fig. 4 to capture v_{ds} patterns swiftly and selectively. By manipulating C_U/C_L , only specific Δv_{ds} can flip its output level (i.e., voltage across C_L , denoted as v_{CL}) from 0 to V_{cc} or V_{cc} to 0, in which way Δv_{ds} is scaled down to digital-like signals. For Δv_{ds} smaller than the desired value (e.g., the switching noise), the output will not be changed. Moreover, considering the fluctuation in the bus voltage, a pull-up or -down resistor (R_U and R_L) is introduced to bypass the low-frequency components in v_{ds} and also reset the status of v_{CL} .

Circuit operation of the CDU is basically illustrated in Fig. 5. When $dv_{ds}/dt > 0$, C_L is charged, v_{CL} increases until $v_{CL} \approx$

V_{cc} , and then v_{CL} is clamped through D_U . If sizing $C_L \gg C_U$, dv_{CL}/dt will be too small to forward bias D_U , i.e., v_{CL} will simply be proportional to Δv_{ds} without being clamped. When $dv_{ds}/dt < 0$, C_L is discharged, v_{CL} decreases until around 0, and then it is clamped via D_L . Therefore, the principle of sizing C_L and C_U is derived as follows.

$$C_L = C_U \left(\frac{\Delta v_{ds}}{V_{cc}} - 1 \right) \quad (1)$$

Here, Δv_{ds} indicates the sensitivity of logic flipping. For instance, $\Delta v_{ds} = 50$ V designs a CDU, the output of which will be clamped at V_{cc} when the input v_{ds} experiences a variation above 50 V. The larger Δv_{ds} in Eq. (1) is set, the lower sensitivity of the designed CDU will be. Combining CDUs of different sensitivities, the fault patterns could be extracted with switching noise immunity achieved. Therefore, according to the fault characteristics analysis in Section II, three CDUs are specifically designed to capture the fault patterns occurring at switching transition, ON- and OFF-states. Denote the three CDUs as $CDU_{(SW)}$, $CDU_{(ON)}$ and $CDU_{(OFF)}$ respectively, and their sensitivities are designed as follows:

- 1) The Δv_{ds} of $CDU_{(SW)}$ should be around DC bus voltage in such way that its logic flip occurs only at switching transition, which refrains the interference of switching noise. Fault patterns (a), (b) and (d) will be detected if $CDU_{(SW)}$ doesn't generate a flipped output during switching transition.
- 2) The Δv_{ds} of $CDU_{(ON)}$ and $CDU_{(OFF)}$ should be set corresponding to the expected Δv_{ds} as shown in Fig. 2 at FUL and HSF respectively. Fault patterns (c), (e) and (f) will be captured when $CDU_{(ON)}$ and $CDU_{(OFF)}$ generate a flipped output during ON- and OFF-states.

In the practical design, it is suggested to firstly determine C_U , and then calculate C_L with Eq. (1). To avoid the CDU contributing to the output capacitance C_{oss} of the DUT, C_U ought to be sufficiently smaller than C_{oss} as it has to withstand almost the entire v_{ds} especially in the high-voltage scenario. In this paper, Eq. (2) is recommended for sizing C_U . For GaN HEMTs, C_U is usually on the order of pF to avoid impacting the switching speed.

$$C_U \sim \frac{C_{oss}}{100} \quad (2)$$

A relatively large capacitor C_0 is implemented on the supply side to stabilize V_{cc} and mitigate the voltage spike in v_{CL} . Thus, C_0 should be way larger than C_L .

On top of the capacitors and diodes, a pull-down or -up resistor (R_L or R_U) is introduced to reset the CDU before its functioning. In $CDU_{(SW)}$, a sufficiently large $R_{L(SW)}$ is used to i) fully discharge $C_{L(SW)}$ before the circuit start-up and ii) avoid significant attenuation in v_{CL} during operation. Thus, $R_{L(SW)}$ for $CDU_{(SW)}$ should meet the following equation:

$$R_{L(SW)} \gg \frac{1 - D_{min}}{C_{L(SW)} f_s} \quad (3)$$

Here, D_{min} and f_s are the minimum duty ratio and switching frequency of PWM apply on the phase leg. Similarly, in $CDU_{(ON)}$, a $R_{L(ON)}$ is also implemented to discharge $C_{L(ON)}$.

Such a pull-down resistor should be relatively smaller since it purposefully resets v_{CL} once a logic flip occurs, making $CDU_{(ON)}$ capture the FUL pattern exclusively. For instance, after the turn-off transition where $dv_{ds}/dt > 0$, v_{CL} of $CDU_{(ON)}$ inevitably flips from "0" to "1" due to its high sensitivity, which is undesired since $CDU_{(ON)}$ is for capturing the fault pattern at ON-state. Hence, $R_{L(ON)}$ should act to immediately pull down v_{CL} from "1" back to "0" to avoid mis-trigger during normal operation. Besides, the impact of $R_{L(ON)}$ on the impedance of $C_{L(ON)}$ needs to be negligible. Thus, the design of $R_{L(ON)}$ has to meet the following equation:

$$R_{L(ON)} \gg \frac{1}{2\pi f_{eff}(C_U + C_{L(ON)})} \quad (4)$$

$$R_{L(ON)} \sim \frac{t_{reset}}{C_{L(ON)}}$$

where

$$f_{eff} \approx \frac{0.35}{\min(t_r, t_f)} \quad (5)$$

Here, t_{reset} is the expected interval to reset v_{CL} after its logic flip, t_r and t_f are respectively the rising and falling time of the DUT's switching action. In $CDU_{(OFF)}$, a pull-up resistor $R_{U(OFF)}$ is used instead, making it exclusively capture the HSF fault at OFF-state. The design consideration for $R_{U(OFF)}$ follows the same principles of that for $R_{L(ON)}$. Therefore, $R_{U(OFF)}$ could be determined by Eqs. (4) and (5) as well.

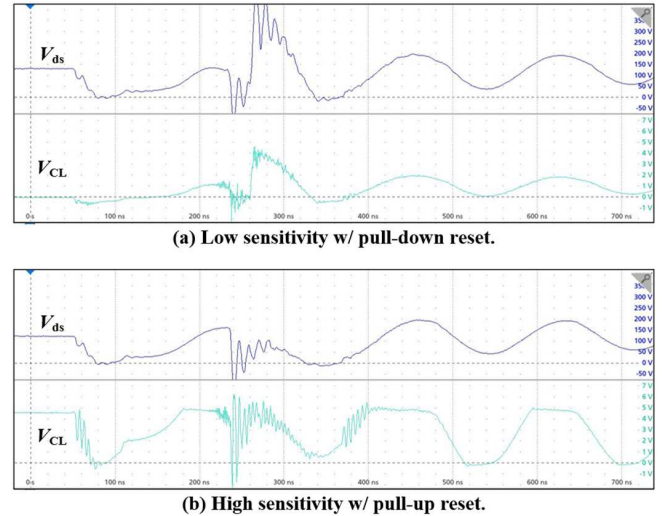


Fig. 6 Response of CDUs of (a) low sensitivity with a pull-down resistor and (b) high sensitivity with a pull-up resistor.

After C_0 absorbs the voltage spike in v_{CL} , the implemented protection diode D_0 , however, blocks its discharging path, making it unable to restore the voltage back to V_{cc} . Thus, R_0 is introduced to intentionally form a discharging path for C_0 . The design principles of R_0 are:

- 1) Being able to dissipate the net charge induced by CDU voltage spike within each switching cycle.
- 2) Merely a small leakage path for V_{cc} .
- 3) Leaving negligible impact on the impedance of C_0 .

Hence, the design of R_0 is depicted by the following equation:

$$\max\left(\frac{1}{2\pi f_{\text{eff}} C_0}, \frac{V_{cc}}{I_{\text{leak}}}\right) \ll R_0 < \frac{V_{cc} - V_F}{f_s [C_U V_{dc} - (C_L + C_U) V_{cc}]} \quad (6)$$

Here, I_{leak} is the allowable leakage current of V_{cc} supply determined by its datasheet, V_F is the forward voltage of D_0 , and V_{dc} is the DC bus voltage.

In Fig. 6, operation of two CDUs with different sensitivity is demonstrated. The CDU waveforms in Fig. 6 (a) features $C_U/C_L = 5 \text{ pF} / 390 \text{ pF}$ with a pull-down resistor of $500 \text{ M}\Omega$, and the other one in Fig. 6 (b) features $C_U/C_L = 5 \text{ pF} / 47 \text{ pF}$ with a pull-up resistor of $2 \text{ k}\Omega$. The low-sensitivity CDU exhibits a well synchronously scaled-down waveform of v_{ds} , while the high-sensitivity part generates a clamped output with the upper and lower boundaries of $\sim 5 \text{ V}$ and $\sim 0 \text{ V}$.

B. Generation of Fault Flag Through Logic Synthesis

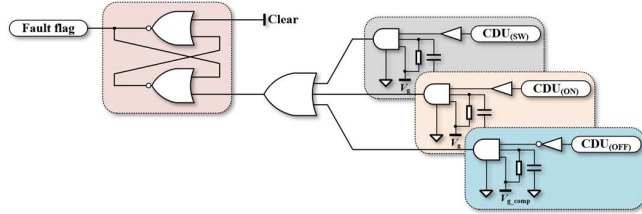


Fig. 7. Digital processing circuit for fault recognition.

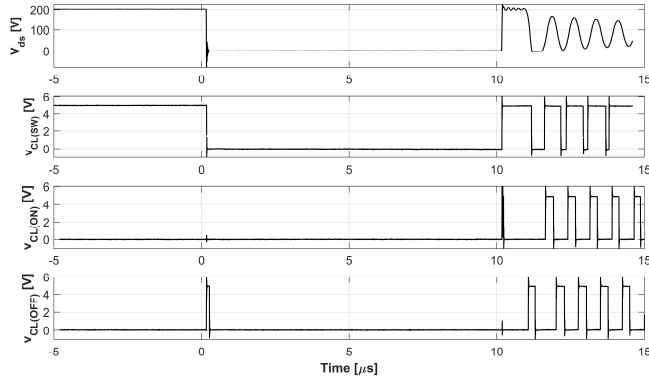


Fig. 8. Test waveforms regarding the digitalization of CDU output.

As discussed in Section II, once the v_{ds} patterns are captured by CDUs, a digital circuit is needed for signal processing with the designed logic synthesis to distinguish the fault from normal operation. The objectives of the logic synthesis include: i) digitalization of v_{ds} patterns, ii) elimination of the switching noise interference in v_{ds} patterns, and iii) recognition of abnormal signal and generation of fault flag. Based on the analysis in Fig. 3 and TABLE 1, the digital processing circuit is proposed and illustrated in Fig. 7. Overall, the circuit comprises four functional layers as follows.

- 1) **Pattern digitalization:** The captured v_{ds} patterns are converted into different v_{CL} by $CDU_{(SW)}$, $CDU_{(ON)}$ and $CDU_{(OFF)}$ (denoted as $v_{CL(SW)}$, $v_{CL(ON)}$ and $v_{CL(OFF)}$). Then, $v_{CL(SW)}$ and $v_{CL(ON)}$ are passed to non-inverting buffer gates for digitalization. Since $CDU_{(SW)}$ has a relatively small sensitivity, its cascaded buffer gate

does not flip the logic only if the switching action occurs, by which the switching noise is bypassed. Meanwhile, as $CDU_{(ON)}$ has a high sensitivity, its cascaded buffer gate experiences logic flip triggered by either FUL or switching noise. The $CDU_{(OFF)}$ has a similar situation, but it is cascaded with a NOT gate to invert $v_{CL(OFF)}$ due to its pull-up resistor that makes its default state “1”. Test results of v_{ds} , $v_{CL(SW)}$, $v_{CL(ON)}$ and $v_{CL(OFF)}$ are demonstrated in Fig. 8 to showcase the functionality. As expected, $v_{CL(SW)}$ is synchronous with v_{ds} , while $v_{CL(ON)}$ and $v_{CL(OFF)}$ only respond to the rising and falling edge of v_{ds} , respectively.

- 2) **Blanking insertion:** Each CDU output has a fault-recognizing submodule consisting of an AND gate with a RC branch and a buffer/inverter. The digitalized v_{CL} is passed to all submodules. Simultaneously, in the submodules for $CDU_{(SW)}$ and $CDU_{(ON)}$, the PWM signals are fed directly, and through a RC branch as well, to the AND gate. Thus, the RC time constant acts as the blanking time to activate the submodule. When the PWM signal is ON, the $CDU_{(ON)}$ submodule is set to be ready for FUL detection. When the PWM is turned off, the AND gate is immediately deactivated without blanking time, which avoids the interference of turn-off noise in v_{ds} . Meanwhile, the $CDU_{(OFF)}$'s submodule will become activated after its blanking time to take over the detection for HSF. Moreover, to decouple the SC fault and switching noise in the captured v_{ds} pattern, the blanking time of $CDU_{(SW)}$ submodule is designed intentionally shorter than that of submodules for $CDU_{(ON)}$ and $CDU_{(OFF)}$, in which way the switching noise fully attenuates before the activation of $CDU_{(ON)}$ and $CDU_{(OFF)}$, while $CDU_{(SW)}$ is able to detect the fault such as patterns (a), (b) and (d). One practical example is the blanking time of submodules for $CDU_{(SW)}$, $CDU_{(ON)}$ and $CDU_{(OFF)}$ are respectively 30 ns, 100 ns and 100 ns.
- 3) **Pattern integration:** The switching-noise-free outputs of three submodules are integrated through a OR gate to recognize/distinguish the normal switching transition, FUL and HSF, among which only the 2nd and 3rd scenarios will make the output logic “1”. As the switching noise is bypassed with the dedicatedly introduced difference in CDU sensitivity and blanking time, logic “1” of the OR gate's output convincingly represents the moment when a SC fault is detected.
- 4) **Flag generation:** Finally, a latch-up built with two NOR gates is implemented to generate a fault flag, which will be then passed through a digital isolator for galvanic isolation during the test. The initial state of the latch-up is “0”, and once it is triggered by the OR gate, it immediately acts to outputs constantly “1”, namely the fault flag.

Overall, the detection speed is determined by i) charging speed of CDUs, ii) propagation delay from buffer/inverter, AND gate, OR gate, latch-up. The selection of ICs considers prioritize the small propagation delay and small input capacitance to improve the detection speed as much as possible.

IV. DESIGN VALIDATION

A. Functional Validation in Circuit Simulation

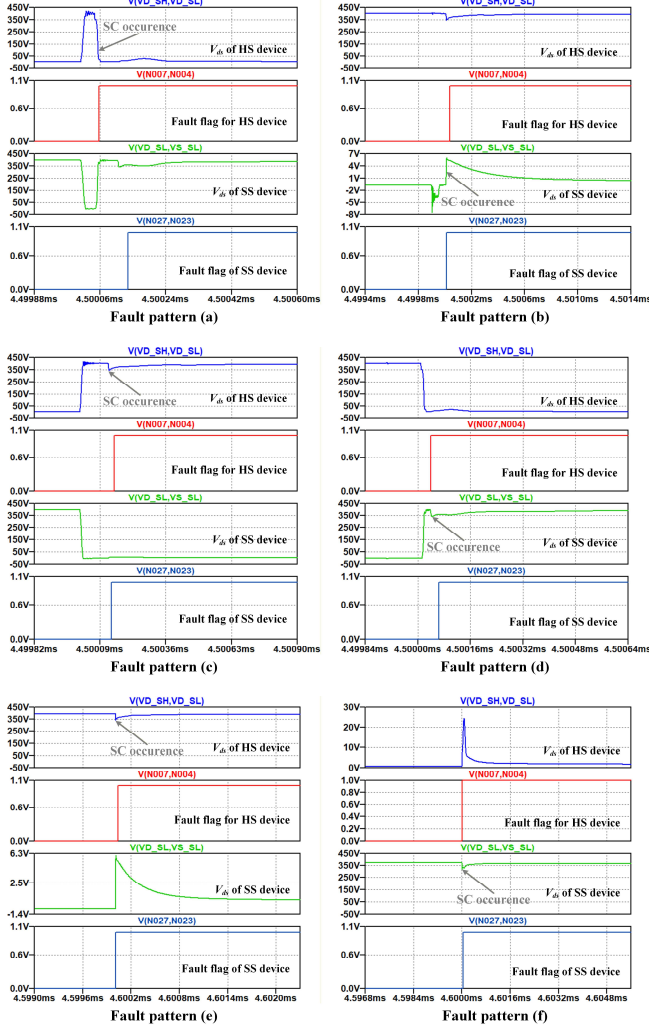


Fig. 9. Functional validation of proposed detection scheme.

A phase leg implemented with the proposed SC detection is simulated in Ltspice, targeting the identical fault scenarios to that in Fig. 3, and the results are shown in Fig. 9. Each simulated patterns includes the waveforms of upper and lower devices' v_{ds} and fault flags generated by corresponding SC detection circuits. The simulation reveals that the proposed detection scheme is adaptive to different types of SC fault regardless of the moment of occurrence. Moreover, it is capable of recognizing the small fault pattern from large v_{ds} while immune to the interference of switching noise.

B. Detection Speed Test With Hardware Prototype

To test the detection speed in hardware implementation, an evaluation board has been built, as shown in Fig. 10. The board comprises i) a phase leg consisting of two GaN HEMTs (GS-065-018-2-L), ii) isolated gate driver for each DUT, and iii) SC detection module for each DUT. Detailed PCB layout and dimensions of the detection module is also illustrated in Fig. 10.

The CDUs built with 0402 NP0-type ceramic capacitors make up totally $5 \times 9 \text{ mm}^2$, and the logic synthesis part makes up $15 \times 9 \text{ mm}^2$, which could be further optimized by using more compact layout and ICs with smaller package.

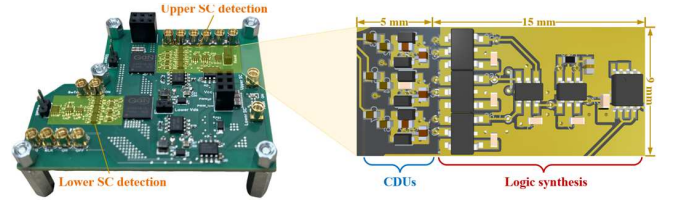


Fig. 10. Prototype of a GaN-HEMT phase leg implemented with SC detection.

To validate the design functionality and robustness, firstly, based on the previously developed testing bench in [16], the double-pulse test (DPT) is performed with a 200-V DC bus voltage to mimic the normal operation scenario. The measured waveforms are presented in Fig. 11. During normal operation, it turns out that the fault flag always remains "0", indicating the detection circuit is robust against the mis-trigger issue induced from either switching transition or switching noise.

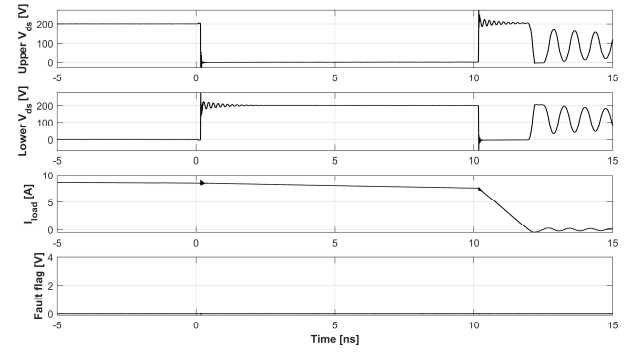


Fig. 11. Waveforms of DPT results, including v_{ds} of the upper and lower DUTs the load current, and the fault flag signal from the lower SC detection.

Then, SC faults are intentionally introduced in the operation. The designed parameters of CDUs are presented in TABLE 2. For the lower DUT, a HSF is created by purposefully pulling up the lower gate voltage when the upper DUT is ON, and a FUL is introduced by pulling up the upper gate voltage when the lower DUT is ON. The detection response time is determined by the difference between the moments of the fault occurrence and fault flag generation for the lower DUT. In Fig. 12, the detection circuit is tested with a HSF and the corresponding fault flag is observed after 32 ns. In Fig. 13, a FUL is introduced and the fault flag is then observed after 35 ns. Considering that there is a digital isolator (SI86S621BC-IS [15]) implemented between latch-up and probe for galvanic isolation, the additional propagation delay from the isolator (13.5 ns) should be excluded from the detection time. Hence, the actual detection time for HSF and FUL is 18.5 ns and 21.5 ns, respectively, validating the feature of ultrafast detection speed of the proposed scheme.

TABLE 2 CDU DESIGN FOR THE SC TEST.

	CDU _(SW)	CDU _(OFF)	CDU _(ON)
C_U [pF]	5	5	5
C_L [pF]	390	47	10

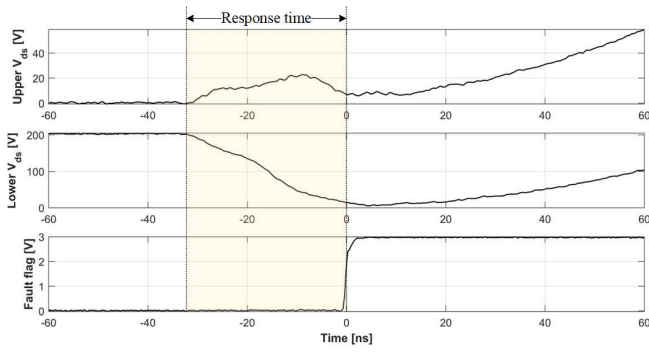


Fig. 12. Detection response delay measurement for HSF.

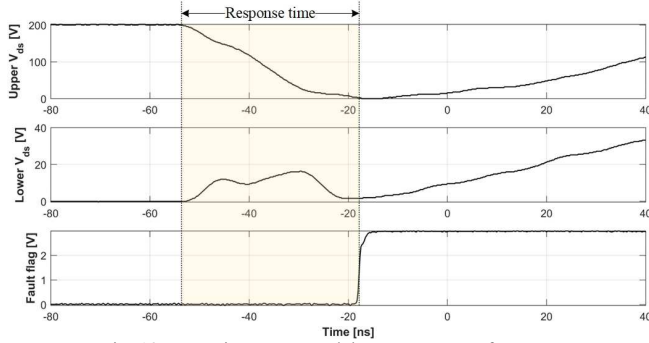


Fig. 13. Detection response delay measurement for FUL.

V. CONCLUSIONS

This paper proposes a novel SC detection scheme based on the fault pattern recognition from entire v_{ds} trail, achieving ultrafast fault detection with switching noise immunity. The analysis on v_{ds} pattern reveals characteristics of SC fault in terms of Δv_{ds} and dv/dt , which distinguishes the fault from normal operation with parasitic ringing. The proposed CDU design considers the uniqueness of fault patterns. It can swiftly extract the key characteristics of the patterns and convert them into low voltage signals for digitalization. With the digitalized v_{ds} patterns, the logic synthesis is developed to recognize the SC fault while bypass the interference of switching noise in v_{ds} , featuring ultrafast detection response speed. The simulation result indicates that the proposed scheme can well handle both HSF and FUL in different scenarios. The SC hardware test result shows that the detection circuit is robust against the mis-trigger issue induced by switching noise during normal operation. The ultrafast detection is validated with the observed response time of 32 ns and 35 ns and actual response time of 18.5 ns and 21.5 ns for HSF and FUL, respectively.

ACKNOWLEDGMENT

This material is based on work supported by NASA under Award No. 80NSSC24K0179.

REFERENCES

- [1] E. A. Jones, "Characterization methodology, modeling, and converter design for 600 V enhancement-mode GaN FETs," 2018.
- [2] Y. Gui et al., "Desaturation detection for paralleled GaN e-HEMT phase leg," in Proc. IEEE Energy Convers. Cong. Expo., 2018, pp. 3503–3509.
- [3] A. K. Das et al., "Investigating the limitations of Desat protection for WBG Semiconductors: conjoint effects due to the parasitic inductances, junction temperature and dynamic drain-source resistance," 2025 IEEE 34th Microelectronics Design & Test Symposium (MDTS), Albany, NY, USA, 2025.
- [4] Infineon. [Online]. Available: <https://www.infineon.com/assets/row/public/documents/24/49/infineon-gs66516t-datasheet-en.pdf?fileId=8ac78c8c8d2fe47b018e5161d33153ea>
- [5] R. Hou et al., "A simple desaturation-based protection circuit for GaN HEMT with ultrafast response," in IEEE Transactions on Power Electronics, vol. 36, no. 6, pp. 6978–6987, June 2021.
- [6] Z. Xin et al., "A Review of Megahertz Current Sensors for Megahertz Power Converters," in IEEE Transactions on Power Electronics, vol. 37, no. 6, pp. 6720–6738, June 2022.
- [7] T. Horiguchi et al., "A short circuit protection method based on a gate charge characteristic," 2014 International Power Electronics Conference (IPEC-Hiroshima 2014 - ECCE ASIA), Hiroshima, Japan, 2014, pp. 2290–2296.
- [8] T. Horiguchi et al., "Short-circuit protection method based on a gate charge characteristic," in IEEE Journal of Industry Applications, vol. 4, no. 4, pp. 360–369, 2015.
- [9] E. Martano et al., "A simple and low cost overcurrent protection system based on commercial shunt for wide-bandgap devices," in PCIM Europe 2024; International Exhibition and Conference for Power Electronics, Intelligent Motion, Renewable Energy and Energy Management, 2024, pp. 2869–2874.
- [10] Ziegler, R. C. Woodward et al., "Current sensing techniques: A review," in IEEE Sensors Journal, vol. 9, no. 4, pp. 354–376, 2009.
- [11] E. R. Motto et al., "IGBT module with user accessible on-chip current and temperature sensors," 2012 Twenty-Seventh Annual IEEE Applied Power Electronics Conference and Exposition (APEC), Orlando, FL, USA, 2012, pp. 176–181.
- [12] J. Le Leslé et al., "A fast short-circuit detection and protection method for wide band-gap devices based on current derivative sensing," in 2021 23rd European Conference on Power Electronics and Applications (EPE'21 ECCE Europe), 2021, pp. 1–9.
- [13] A. Mutlu et al., "Advantages of rogers coil over desaturation method for leg short circuit detection in inverters," in PCIM Europe 2022; International Exhibition and Conference for Power Electronics, Intelligent Motion, Renewable Energy and Energy Management, 2022, pp. 1–6.
- [14] P. T. N. Kishore et al., "Analytical modelling of fault transient in a GaN HEMT half bridge and its overcurrent protection with PCB embedded rogers coils," in IEEE Transactions on Industry Applications, pp. 1–11, 2024.
- [15] Skyworks solutions. [Online]. Available: https://www.skyworksinc.com/-/media/SkyWorks/SL/documents/public/data-sheets/si86s61x_2x_Ox-datasheet.pdf
- [16] T. Qiu, Z. Zhang, P. Khadka, A. Siraj and D. Rana, "Automated Characterization Platform for Comprehensive Dynamic Rds(on) Assessment of GaN HEMTs From 50 K to 400 K," 2025 IEEE Applied Power Electronics Conference and Exposition (APEC), Atlanta, GA, USA, 2025, pp. 1040–1046.